Chapter 10. Applications of Theta Functions

Jacobi triple product, transformation laws, partitions, the two- and four-squares theorems

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Overview

The *theta function*

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}$$

converges for all $z \in \mathbb{C}$ and τ in the upper half-plane. It has a dual nature: as a function of z it belongs to the world of elliptic functions (period 1, "quasi-period" τ); as a function of τ it is modular and connects with partitions and with representing integers as sums of squares. The main tools are the *triple product formula* and the *transformation law*. We then discuss partitions and prove the celebrated two-squares and four-squares theorems.

1 Product formula for the Jacobi theta function

Motivation. The theta function $\Theta(z|\tau)$ is an entire function with explicit zeros, and its Jacobi product formula is the source of many identities in number theory, including the generating functions and the sums of squares theorems at the end of the chapter.

For $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$, **Jacobi's theta function** is

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}. \tag{1}$$

Special cases: $\theta(\tau) = \sum_n e^{\pi i n^2 \tau} = \Theta(0|\tau)$ for $\tau \in \mathbb{H}$, and $\vartheta(t) = \sum_n e^{-\pi n^2 t} = \theta(it)$ for $t > 0$. The heat kernel on the circle (Book I) is $H_t(x) = \sum_n e^{-4\pi^2 n^2 t} e^{2\pi i n x} = \Theta(x|4\pi i t)$; and the functional equation of ϑ gave that of ζ (Chapter 6).

Proposition 1.1

The function Θ satisfies the following properties:
(i) Θ is entire in $z \in \mathbb{C}$ and holomorphic in $\tau \in \mathbb{H}$.
(ii) $\Theta(z+1|\tau) = \Theta(z|\tau)$.
(iii) $\Theta(z+\tau|\tau) = \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}$.
(iv) $\Theta(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$.

Proof. If $\mathrm{Im}(\tau) = t \geq t_0 > 0$ and $|z| \leq M$, then $\sum_n |e^{\pi i n^2 \tau} e^{2\pi i n z}| \leq C \sum_{n \geq 0} e^{-\pi n^2 t_0} e^{2\pi n M} < \infty$, uniformly; this gives (i).

(ii) is clear since $e^{2\pi i n z}$ has period 1. For (iii), complete the square:

$$\Theta(z+\tau|\tau) = \sum_n e^{\pi i (n^2+2n)\tau} e^{2\pi i n z} = \sum_n e^{\pi i (n+1)^2 \tau} e^{-\pi i \tau} e^{2\pi i (n+1) z} e^{-2\pi i z} = \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}.$$

By (ii), (iii) it suffices for (iv) to show $\Theta(1/2 + \tau/2|\tau) = 0$: $\Theta(1/2+\tau/2|\tau) = \sum_n (-1)^n e^{\pi i (n^2+n)\tau}$, and pairing $n \geq 0$ with $-n-1$, which has opposite parity and $(-n-1)^2 + (-n-1) = n^2 + n$, the terms cancel. ∎

With the standard notation $q = e^{\pi i \tau}$, the **triple product** is

$$\Pi(z|\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1} e^{2\pi i z})(1 + q^{2n-1} e^{-2\pi i z}).$$

Proposition 1.2

The function $\Pi(z|\tau)$ satisfies the following properties:
(i) $\Pi(z|\tau)$ is entire in $z \in \mathbb{C}$ and holomorphic for $\tau \in \mathbb{H}$.
(ii) $\Pi(z+1|\tau) = \Pi(z|\tau)$.
(iii) $\Pi(z+\tau|\tau) = \Pi(z|\tau)e^{-\pi i\tau}e^{-2\pi iz}$.
(iv) $\Pi(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$. Moreover, these points are simple zeros of $\Pi(\cdot|\tau)$, and $\Pi(\cdot|\tau)$ has no other zeros.

Proof. If $\mathrm{Im}(\tau) \geq t_0 > 0$, $|q| \leq e^{-\pi t_0} < 1$ and each factor is $1+O(|q|^{2n-1}e^{2\pi|z|})$; since $\sum|q|^{2n-1} < \infty$, the results on infinite products (Section 3 of Chapter 5) give (i).

(ii) is clear. For (iii), since $q^2 = e^{2\pi i\tau}$,

$$\Pi(z+\tau|\tau) = \prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n+1}e^{2\pi iz})(1+q^{2n-3}e^{-2\pi iz}),$$

and comparing with $\Pi(z|\tau)$, $\Pi(z+\tau|\tau) = \Pi(z|\tau)\frac{1+q^{-1}e^{-2\pi iz}}{1+qe^{2\pi iz}}$;

(iii) follows from $(1+x)/(1+x^{-1}) = x$ for $x \neq -1$. For (iv), a convergent product vanishes only if a factor does. $1-q^{2n} \neq 0$. $1+q^{2n-1}e^{2\pi iz} = 0$ iff $q^{2n-1}e^{2\pi iz} = -1 = e^{\pi i}$, i.e. $(2n-1)\tau + 2z \equiv 1 \pmod 2$, i.e. $z \equiv 1/2 + \tau/2 - n\tau \pmod 1$ $(n \geq 1)$. $1+q^{2n-1}e^{-2\pi iz} = 0$ iff $(2n-1)\tau - 2z \equiv 1 \pmod 2$, i.e. $z \equiv -1/2 - \tau/2 + n\tau \equiv 1/2 + \tau/2 + (n-1)\tau \pmod 1$ $(n \geq 1)$. This exhausts the zeros, and they are simple since $e^w - 1$ vanishes to order 1 at 0. ∎

Theorem 1.3 (Product formula)

For all $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ we have the identity $\Theta(z|\tau) = \Pi(z|\tau)$.

Proof. Fix $\tau \in \mathbb{H}$. There is a constant $c(\tau)$ with

$$\Theta(z|\tau) = c(\tau)\Pi(z|\tau)\,: \tag{2}$$

$F(z) = \Theta(z|\tau)/\Pi(z|\tau)$ is entire (by (iv) of both propositions) and doubly periodic with periods 1, τ (by (ii), (iii)), hence constant. We show $c(\tau) = c(4\tau)$. Put $z = 1/2$ in (2) ($e^{\pm 2\pi iz} = -1$):

$$\sum_n (-1)^n q^{n^2} = c(\tau)\prod_{n=1}^{\infty}(1-q^{2n})(1-q^{2n-1})^2 = c(\tau)\prod_{n=1}^{\infty}(1-q^n)(1-q^{2n-1}), \quad c(\tau) = \frac{\sum_n(-1)^n q^{n^2}}{\prod_{n=1}^{\infty}(1-q^n)(1-q^{2n-1})}. \tag{3}$$

Put $z = 1/4$ ($e^{2\pi iz} = i$): $\Theta(1/4|\tau) = \sum_n q^{n^2}i^n$, and since $1/i = -i$, only even $n = 2m$ survive: $\Theta(1/4|\tau) = \sum_m q^{4m^2}(-1)^m$. On the other hand

$$\Pi(1/4|\tau) = \prod_{m=1}^{\infty}(1-q^{2m})(1+iq^{2m-1})(1-iq^{2m-1}) = \prod_{m=1}^{\infty}(1-q^{2m})(1+q^{4m-2}) = \prod_{n=1}^{\infty}(1-q^{4n})(1-q^{8n-4}),$$

the last step by splitting $2m = 4n$ and $2m = 4n-2$ in the first factor ($(1-q^{4n-2})(1+q^{4n-2}) = 1-q^{8n-4}$). Hence

$$c(\tau) = \frac{\sum_n(-1)^n q^{4n^2}}{\prod_{n=1}^{\infty}(1-q^{4n})(1-q^{8n-4})}, \tag{4}$$

and comparing (3) and (4) ($q \mapsto q^4$ corresponds to $\tau \mapsto 4\tau$), $c(\tau) = c(4\tau)$. So $c(\tau) = c(4^k\tau)$, and since $q^{4^k} = e^{\pi i 4^k\tau} \to 0$, (3) gives $c(4^k\tau) \to 1$; so $c(\tau) = 1$. ∎

Corollary 1.4

If $\mathrm{Im}(\tau) > 0$ and $q = e^{\pi i \tau}$, then

$$\theta(\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1})^2.$$

Thus $\theta(\tau) \neq 0$ for $\tau \in \mathbb{H}$.

Corollary 1.5

For each fixed $\tau \in \mathbb{H}$, the quotient

$$(\log \Theta(z|\tau))'' = \frac{\Theta(z|\tau)\Theta''(z|\tau) - (\Theta'(z|\tau))^2}{\Theta(z|\tau)^2}$$

is an elliptic function of order 2 with periods 1 and τ, and with a double pole at $z = 1/2 + \tau/2$.

(Primes denote differentiation in z.)

Proof. Let $F = (\log \Theta)' = \Theta'/\Theta$. Differentiating (ii), (iii) of Proposition 1.1 gives $F(z+1) = F(z)$ and $F(z+\tau) = F(z) - 2\pi i$, so F' is doubly periodic. Θ vanishes in the fundamental parallelogram only at $1/2 + \tau/2$, simply, so F has only a simple pole there and F' only a double pole. ∎

1.1 Further transformation laws

In Chapter 9 the modular character was expressed through $T_1 : \tau \mapsto \tau + 1$ and $S : \tau \mapsto -1/\tau$. For Θ it is natural to use $T_2 : \tau \mapsto \tau + 2$ and S, since $\Theta(z|\tau + 2) = \Theta(z|\tau)$ but $\Theta(z|\tau + 1) \neq \Theta(z|\tau)$.

Theorem 1.6

If $\tau \in \mathbb{H}$, then

$$\Theta(z|-1/\tau) = \sqrt{\tau/i}\, e^{\pi i z^2 \tau} \Theta(z\tau|\tau) \qquad \text{for all } z \in \mathbb{C}. \tag{5}$$

Here $\sqrt{\tau/i}$ denotes the branch of the square root defined on the upper half-plane that is positive when $\tau = it$, $t > 0$.

Proof. It suffices to prove (5) for $z = x$ real and $\tau = it$, $t > 0$: for fixed real x both sides are holomorphic in $\tau \in \mathbb{H}$ and would agree on the positive imaginary axis; and for fixed τ both sides are entire in z and would agree on $\mathbb{R}$. For $z = x$, $\tau = it$, (5) reads

$$\sum_n e^{-\pi n^2/t} e^{2\pi i n x} = t^{1/2} e^{-\pi t x^2} \sum_n e^{-\pi n^2 t} e^{-2\pi n x t}.$$

Replacing x by a, this is $\sum_n e^{-\pi t(n+a)^2} = \sum_n t^{-1/2} e^{-\pi n^2/t} e^{2\pi i n a}$, which is equation (3) of Chapter 4 (from Poisson summation). ∎

Corollary 1.7

If $\mathrm{Im}(\tau) > 0$, then $\theta(-1/\tau) = \sqrt{\tau/i}\, \theta(\tau)$.

(Set $z = 0$; for $\tau = it$ this is the functional equation of ϑ.) This gives precise information about θ as $\tau \to 0$; the next corollary describes θ near $\tau = 1$.

Corollary 1.8

If $\tau \in \mathbb{H}$, then

$$\theta(1-1/\tau) = \sqrt{\tau/i} \sum_{n=-\infty}^{\infty} e^{\pi i(n+1/2)^2\tau} = \sqrt{\tau/i}\,(2e^{\pi i\tau/4} + \cdots).$$

The second identity means that $\theta(1-1/\tau) \sim 2\sqrt{\tau/i}\, e^{\pi i\tau/4}$ as $\mathrm{Im}(\tau) \to \infty$.

Proof. n and n^2 have the same parity, so $\theta(1+\tau) = \sum_n (-1)^n e^{\pi i n^2\tau} = \Theta(1/2|\tau)$; hence $\theta(1-1/\tau) = \Theta(1/2|-1/\tau)$. By Theorem 1.6 with $z = 1/2$,

$$\theta(1-1/\tau) = \sqrt{\tau/i}\, e^{\pi i\tau/4}\Theta(\tau/2|\tau) = \sqrt{\tau/i}\, e^{\pi i\tau/4} \sum_n e^{\pi i n^2\tau} e^{\pi i n\tau} = \sqrt{\tau/i} \sum_n e^{\pi i(n+1/2)^2\tau}.$$

The terms $n = 0, -1$ contribute $2e^{\pi i\tau/4}$, of absolute value $2e^{-\pi t/4}$ ($\tau = \sigma + it$); the others total $O(\sum_{k\geq 1} e^{-\pi(k+1/2)^2 t}) = O(e^{-9\pi t/4})$. ∎

The **Dedekind eta function** is, for $\mathrm{Im}(\tau) > 0$,

$$\eta(\tau) = e^{\frac{\pi i\tau}{12}} \prod_{n=1}^{\infty} (1 - e^{2\pi i n\tau}).$$

Proposition 1.9

If $\mathrm{Im}(\tau) > 0$, then $\eta(-1/\tau) = \sqrt{\tau/i}\,\eta(\tau)$.

Proof. Differentiate (5) and evaluate at $z_0 = 1/2 + \tau/2$. By the product formula,

$$\Theta(z|\tau) = (1 + qe^{-2\pi iz}) \prod_{n=1}^{\infty} (1-q^{2n})(1+q^{2n-1}e^{2\pi iz})(1+q^{2n+1}e^{-2\pi iz}),$$

and the first factor vanishes at z_0 (where $e^{2\pi iz_0} = -q$), with derivative $-2\pi iqe^{-2\pi iz_0} = 2\pi i$ there, while the remaining factors equal $(1-q^{2n})^3$. So

$$\Theta'(z_0|\tau) = 2\pi i H(\tau), \qquad H(\tau) = \prod_{n=1}^{\infty} (1 - e^{2\pi i n\tau})^3.$$

Replacing $-1/\tau$ by τ in (5), $\Theta(z|\tau) = \sqrt{i/\tau}\, e^{-\pi i z^2/\tau}\Theta(-z/\tau|-1/\tau)$. Differentiating and evaluating at z_0 (where $-z_0/\tau = -\frac{1}{2\tau} - \frac{1}{2}$ is a zero of $\Theta(\cdot|-1/\tau)$, congruent to $\frac{1}{2} + \frac{-1/\tau}{2}$), a computation gives

$$2\pi i H(\tau) = \sqrt{i/\tau}\, e^{-\frac{\pi i\tau}{4}} e^{-\frac{\pi i}{2}} e^{-\frac{\pi i}{4\tau}} \Big(-\frac{1}{\tau}\Big) 2\pi i H(-1/\tau), \qquad \text{i.e.} \qquad e^{\frac{\pi i\tau}{4}} H(\tau) = \Big(\frac{i}{\tau}\Big)^{3/2} e^{-\frac{\pi i}{4\tau}} H(-1/\tau),$$

which says $\eta(\tau)^3 = (i/\tau)^{3/2}\eta(-1/\tau)^3$. For $\tau = it$, $\eta(\tau) > 0$, so taking cube roots gives $\eta(\tau) = \sqrt{i/\tau}\,\eta(-1/\tau)$; by analytic continuation this holds on $\mathbb{H}$. ∎

2 Generating functions

For a sequence $\{F_n\}_{n=0}^{\infty}$, its **generating function** is $F(x) = \sum_{n=0}^{\infty} F_n x^n$; properties of the sequence translate into algebraic or analytic properties of F, which can lead back to new insights. The **partition function** $p(n)$ is the number of ways n can be written as a sum of positive integers ($p(0) = 1$; e.g. $p(1) = 1$, $p(2) = 2$, $p(3) = 3$, $p(4) = 5$, $p(5) = 7$, $p(6) = 11$, $p(7) = 15$, $p(8) = 22$, $p(12) = 77$).

Theorem 2.1 (Euler)

If $|x| < 1$, then $\displaystyle\sum_{n=0}^{\infty} p(n)x^n = \prod_{k=1}^{\infty} \frac{1}{1-x^k}$.

Formally, write $\frac{1}{1-x^k} = \sum_{m=0}^{\infty} x^{km}$ and multiply out: grouping equal parts, a partition is $n = m_1k_1 + \cdots + m_rk_r$ with distinct k_j, corresponding to the term $(x^{k_1})^{m_1} \cdots (x^{k_r})^{m_r}$. The justification is as for the product formula for ζ (Section 1 of Chapter 7), using convergence of $\prod 1/(1-x^k)$, since $\frac{1}{1-x^k} = 1 + O(x^k)$.

Similarly $\prod 1/(1-x^{2n-1})$ generates $p_o(n)$, the number of partitions of n into odd parts, and $\prod(1+x^n)$ generates $p_u(n)$, partitions into unequal parts. Remarkably $p_o(n) = p_u(n)$, i.e.

$$\prod_{n=1}^{\infty} \frac{1}{1-x^{2n-1}} = \prod_{n=1}^{\infty} (1+x^n) :$$

indeed $\prod(1+x^n)\prod(1-x^n) = \prod(1-x^{2n})$, and by parity $\prod(1-x^{2n})\prod(1-x^{2n-1}) = \prod(1-x^n)$.

Let $p_{e,u}(n)$, $p_{o,u}(n)$ be the numbers of partitions of n into an even, resp. odd, number of unequal parts. Euler proved $p_{e,u}(n) = p_{o,u}(n)$ unless n is a **pentagonal number** $k(3k+1)/2$, $k \in \mathbb{Z}$ (namely $1, 2, 5, 7, 12, 15, 22, 26, \ldots$), in which case $p_{e,u}(n) - p_{o,u}(n) = (-1)^k$. Since multiplying out gives terms $(-1)^r x^{n_1+\cdots+n_r}$ with distinct n_j,

$$\prod_{n=1}^{\infty}(1-x^n) = \sum_{n=0}^{\infty}[p_{e,u}(n) - p_{o,u}(n)]x^n,$$

so Euler's theorem follows from:

Proposition 2.2 (Euler's pentagonal number theorem)

$$\prod_{n=1}^{\infty}(1-x^n) = \sum_{k=-\infty}^{\infty}(-1)^k x^{\frac{k(3k+1)}{2}}.$$

Proof. Set $x = e^{2\pi iu}$. In the triple product take $q = e^{3\pi iu}$ and $z = 1/2 + u/2$; then

$$\prod_{n=1}^{\infty}(1-e^{2\pi i3nu})(1-e^{2\pi i(3n-1)u})(1-e^{2\pi i(3n-2)u}) = \prod_{n=1}^{\infty}(1-e^{2\pi inu}).$$

By Theorem 1.3 the product equals $\sum_n e^{3\pi in^2u}(-1)^n e^{2\pi inu/2} = \sum_n (-1)^n e^{\pi in(3n+1)u} = \sum_n (-1)^n x^{n(3n+1)/2}$. ∎

The growth of $p(n)$ can be analyzed via $1/\prod(1-x^n)$ as $|x| \to 1$; elementary considerations give a rough order of growth, and the transformation law of η (Proposition 1.9) leads to a precise asymptotic formula (Appendix A).

3 The theorems about sums of squares

Let $r_2(n)$ be the number of pairs $(x, y) \in \mathbb{Z}^2$ with $n = x^2 + y^2$ (e.g. $r_2(3) = 0$, $r_2(5) = 8$). *Sum of two squares:* which integers are sums of two squares, and what is $r_2(n)$? No integer $4k+3$ is a sum of two squares, and no integer $8k+7$ is a sum of three squares. Let $r_4(n)$ be the number of representations of n as a sum of four squares. *Sum of four squares:* is every positive integer a sum of four squares, and what is $r_4(n)$? These problems, going back to the third century, were first fully solved with Jacobi's theta functions.

3.1 The two-squares theorem

If $n = a^2 + b^2$ and $m = c^2 + d^2$, then with $x + iy = (a + ib)(c + id)$, $x^2 + y^2 = nm$; so divisibility properties of n matter. Let $d_1(n)$ and $d_3(n)$ be the numbers of divisors of n of the form $4k+1$ and $4k+3$.

Theorem 3.1 (Two-squares theorem)

If $n \geq 1$, then $r_2(n) = 4(d_1(n) - d_3(n))$.

Consequently, if $n = p_1^{a_1} \cdots p_r^{a_r}$ with distinct primes p_j, then n is a sum of two squares iff every prime p_j of the form $4k+3$ has an even exponent a_j. The starting point: for $q = e^{\pi i \tau}$, $\tau \in \mathbb{H}$,

$$\theta(\tau)^2 = \sum_{n=0}^{\infty} r_2(n) q^n, \tag{6}$$

since $\theta(\tau)^2 = \sum_{n_1} \sum_{n_2} q^{n_1^2} q^{n_2^2} = \sum_{(n_1,n_2)\in\mathbb{Z}^2} q^{n_1^2+n_2^2}$.

Proposition 3.2

The identity $r_2(n) = 4(d_1(n) - d_3(n))$, $n \geq 1$, is equivalent to the identities

$$\theta(\tau)^2 = 2 \sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = 1 + 4 \sum_{n=1}^{\infty} \frac{q^n}{1 + q^{2n}}, \tag{7}$$

whenever $q = e^{\pi i \tau}$ and $\tau \in \mathbb{H}$.

Proof. Both series converge absolutely ($|q| < 1$), and they are equal since $1/(q^n + q^{-n}) = q^{|n|}/(1 + q^{2|n|})$. Since $(1 + q^{2n})^{-1} = (1 - q^{2n})/(1 - q^{4n})$, the right side of (7) is $1 + 4\sum_{n\geq 1} \left[\frac{q^n}{1-q^{4n}} - \frac{q^{3n}}{1-q^{4n}}\right]$. Now $\frac{1}{1-q^{4n}} = \sum_{m\geq 0} q^{4nm}$, so

$$\sum_{n=1}^{\infty} \frac{q^n}{1-q^{4n}} = \sum_{n=1}^{\infty} \sum_{m=0}^{\infty} q^{n(4m+1)} = \sum_{k=1}^{\infty} d_1(k) q^k, \qquad \sum_{n=1}^{\infty} \frac{q^{3n}}{1-q^{4n}} = \sum_{k=1}^{\infty} d_3(k) q^k$$

(the series converge since $d_j(k) \leq k$). Compare with (6). ∎

Let

$$C(\tau) = 2 \sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = \sum_{n=-\infty}^{\infty} \frac{1}{\cos(\pi n \tau)}, \qquad q = e^{\pi i \tau},\ \tau \in \mathbb{H}. \tag{8}$$

We must prove $\theta^2 = C$. There is a striking parallel: θ comes from the heat equation on the line, whose kernel is the Gaussian $e^{-\pi x^2}$, its own Fourier transform, and its transformation law comes from Poisson summation; C comes from the steady-state heat equation in a strip, whose kernel is $1/\cosh \pi x$ (Section 1), again its own Fourier transform (Example 3 in Section 2 of Chapter 3), and its transformation law again comes from Poisson summation. Indeed, $\theta(\tau)^2 = (i/\tau)\theta(-1/\tau)^2$ (Corollary 1.7); and (5) of Chapter 4 with $a = 0$ gives $\sum_n \frac{1}{\cosh \pi n t} = \frac{1}{t} \sum_n \frac{1}{\cosh(\pi n/t)}$, i.e. $C(\tau) = (i/\tau)C(-1/\tau)$ for $\tau = it$, hence for all $\tau \in \mathbb{H}$. Both θ^2 and C tend to 1 as $\mathrm{Im}(\tau) \to \infty$. At the "cusp" $\tau = 1$: by Corollary 1.8, $\theta(1 - 1/\tau)^2 \sim 4(\tau/i)e^{\pi i \tau/2}$; and (5) of Chapter 4 with $a = 1/2$ gives $\sum_n \frac{(-1)^n}{\cosh(\pi n/t)} = t \sum_n \frac{1}{\cosh(\pi(n+1/2)t)}$, so by analytic continuation

$$C(1 - 1/\tau) = \frac{\tau}{i} \sum_n \frac{1}{\cos(\pi(n + 1/2)\tau)} = 4\frac{\tau}{i} e^{\pi i \tau/2} + O(|\tau| e^{-3\pi t/2}) \quad (t \to \infty),$$

the main terms being $n = 0, -1$.

Proposition 3.3

The function $C(\tau) = \sum 1/\cos(\pi n\tau)$, defined in the upper half-plane, satisfies (i) $C(\tau+2) = C(\tau)$.
(ii) $C(\tau) = (i/\tau)C(-1/\tau)$.
(iii) $C(\tau) \to 1$ as $\mathrm{Im}(\tau) \to \infty$.
(iv) $C(1-1/\tau) \sim 4(\tau/i)e^{\pi i\tau/2}$ as $\mathrm{Im}(\tau) \to \infty$. Moreover, $\theta(\tau)^2$ satisfies the same properties.

Theorem 3.4

Suppose f is a holomorphic function in the upper half-plane that satisfies:
(i) $f(\tau+2) = f(\tau)$, (ii) $f(-1/\tau) = f(\tau)$, (iii) $f(\tau)$ is bounded, then f is constant.

Let $\mathcal{F} = \{\tau \in \overline{\mathbb{H}} : |\mathrm{Re}(\tau)| \le 1,\ |\tau| \ge 1\}$; the points $\tau = \pm 1$ are its **cusps** (equivalent under $\tau \mapsto \tau+2$).

Lemma 3.5

Every point in the upper half-plane can be mapped into $\mathcal{F}$ using repeatedly one or another of the following fractional linear transformations or their inverses: $T_2 : \tau \mapsto \tau+2$, $S : \tau \mapsto -1/\tau$.

$\mathcal{F}$ is the **fundamental domain** for the group G generated by T_2 and S (uniqueness of representatives fails only on $\partial\mathcal{F}$). Elements $g \in G$ are represented by matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $g(\tau) = \frac{a\tau+b}{c\tau+d}$; these have integer entries and determinant 1, so

$$\mathrm{Im}(g(\tau)) = \frac{\mathrm{Im}(\tau)}{|c\tau+d|^2}. \tag{9}$$

Proof. For $\tau \in \mathbb{H}$, since c, d are integers, by (9) we may choose $g_0 \in G$ with $\mathrm{Im}(g_0(\tau))$ maximal. Translations $T_2^{\pm 1}$ do not change imaginary parts, so there is $g_1 \in G$ with $|\mathrm{Re}(g_1(\tau))| \le 1$ and $\mathrm{Im}(g_1(\tau))$ maximal. If $|g_1(\tau)| < 1$, then $\mathrm{Im}(Sg_1(\tau)) = \mathrm{Im}(-1/g_1(\tau)) = \mathrm{Im}(g_1(\tau))/|g_1(\tau)|^2 > \mathrm{Im}(g_1(\tau))$, contradicting maximality. So $g_1(\tau) \in \mathcal{F}$. ∎

Proof of Theorem 3.4. Suppose f is not constant. Let $g(z) = f(\tau)$ with $z = e^{\pi i\tau}$; g is well defined in the punctured unit disc (period 2) and bounded near 0, so 0 is removable and $\lim_{\mathrm{Im}\,\tau\to\infty} f(\tau)$ exists; by the maximum modulus principle $\lim_{\mathrm{Im}\,\tau\to\infty}|f(\tau)| < \sup_{\mathcal{F}}|f|$. At the cusps (by $f(\tau+2) = f(\tau)$ it suffices to treat $\tau = 1$) we show $\lim_{\mathrm{Im}\,\tau\to\infty} f(1-1/\tau)$ exists and $\lim|f(1-1/\tau)| < \sup_{\mathcal{F}}|f|$, by interchanging 1 and ∞. Let U_n be the fractional linear transformation of $\begin{pmatrix} 1-n & n \\ -n & 1+n \end{pmatrix}$, which fixes 1; with $\nu(\tau) = 1/(1-\tau)$ (taking 1 to ∞, with inverse $\nu^{-1}(\tau) = 1-1/\tau$), $U_n = \nu^{-1}T_n\nu$, $T_n(\tau) = \tau+n$. Hence $U_nU_m = U_{n+m}$ and $U_{-1} = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} = T_2S$. So every U_n is a finite composite of T_2, S and their inverses, and f is invariant under U_n: $f(\nu^{-1}T_n\nu(\tau)) = f(\tau)$. Thus $F(\tau) = f(\nu^{-1}(\tau)) = f(1-1/\tau)$ has period 1, and as above $h(z) = F(\tau)$, $z = e^{2\pi i\tau}$, has a removable singularity at 0, giving the claim by the maximum principle. Hence f attains its maximum in the interior of $\mathbb{H}$, contradicting the maximum principle. ∎

Proof of the two-squares theorem. $f(\tau) = C(\tau)/\theta(\tau)^2$ is holomorphic in $\mathbb{H}$ ($\theta \neq 0$ by Corollary 1.4) and invariant under T_2 and S (Proposition 3.3). In $\mathcal{F}$ it is bounded, tending to 1 as $\mathrm{Im}(\tau) \to \infty$ or $\tau \to \pm 1$, by (iii), (iv) of Proposition 3.3, verified by both C and θ^2. So f is bounded in $\mathbb{H}$, hence constant by Theorem 3.4, and the constant is 1: $\theta^2 = C$. ∎

3.2 The four-squares theorem

Let $\sigma_1^*(n)$ be the sum of the divisors of n that are not divisible by 4.

Theorem 3.6 (Four-squares theorem)

Every positive integer is the sum of four squares, and moreover $r_4(n) = 8\sigma_1^*(n)$ for all $n \geq 1$.

As before, $\theta(\tau)^4 = \sum_{n\geq 0} r_4(n)q^n$, $q = e^{\pi i\tau}$. Here nothing as simple as C is available; we need a subtle variant of the Eisenstein series. Recall the forbidden Eisenstein series $F(\tau) = \sum_m \sum_n \frac{1}{(n+m\tau)^2}$ (sum first in n, then m; $(n,m) \neq (0,0)$) of Corollary 2.6, and define

$$E_2^*(\tau) = F(\tau/2) - 4F(2\tau), \qquad \tau \in \mathbb{H}, \tag{10}$$

equivalently $E_2^*(\tau) = \sum_m \sum_n \left[\frac{1}{(n+m\tau/2)^2} - \frac{1}{(n/2+m\tau)^2}\right]$; the order of summation is critical.

Proposition 3.7

The assertion $r_4(n) = 8\sigma_1^*(n)$ is equivalent to the identity $\theta(\tau)^4 = -\pi^{-2}E_2^*(\tau)$, where $\tau \in \mathbb{H}$.

Proof. It suffices to show $-\pi^{-2}E_2^*(\tau) = 1 + 8\sum_{k\geq 1}\sigma_1^*(k)q^k$. By Corollary 2.6, $F(\tau) = \frac{\pi^2}{3} - 8\pi^2\sum_{k\geq 1}\sigma_1(k)e^{2\pi ik\tau}$, $\sigma_1(k)$ the sum of divisors of k. Now $\sigma_1^*(n) = \sigma_1(n)$ if $4 \nmid n$, and $\sigma_1^*(n) = \sigma_1(n) - 4\sigma_1(n/4)$ if $4 \mid n$ (a divisor $d = 4\tilde{d}$ of $n = 4\tilde{n}$ corresponds to the divisor $\tilde{d}$ of $\tilde{n}$). With (10),

$$E_2^*(\tau) = -\pi^2 - 8\pi^2\sum_{k=1}^{\infty}\sigma_1^*(k)e^{\pi ik\tau}.$$

■

So Theorem 3.6 reduces to $\theta^4 = -\pi^{-2}E_2^*$, and the key is that E_2^* has the same modular properties as θ^4.

Proposition 3.8

The function $E_2^*(\tau)$ defined in the upper half-plane has the following properties:
(i) $E_2^*(\tau+2) = E_2^*(\tau)$.
(ii) $E_2^*(\tau) = -\tau^{-2}E_2^*(-1/\tau)$.
(iii) $E_2^*(\tau) \to -\pi^2$ as $\mathrm{Im}(\tau) \to \infty$.
(iv) $|E_2^*(1-1/\tau)| = O(|\tau^2 e^{\pi i\tau}|)$ as $\mathrm{Im}(\tau) \to \infty$. Moreover $-\pi^2\theta^4$ has the same properties.

(i) is immediate from the definition. Consider F and its *reverse* $\tilde{F}(\tau) = \sum_n \sum_m \frac{1}{(n+m\tau)^2}$ (summing first in m).

Lemma 3.9

The functions F and $\tilde{F}$ satisfy:
(a) $F(-1/\tau) = \tau^2\tilde{F}(\tau)$, (b) $F(\tau) - \tilde{F}(\tau) = 2\pi i/\tau$, (c) $F(-1/\tau) = \tau^2 F(\tau) - 2\pi i\tau$.

Proof. (a) follows from the definitions and $(n+m(-1/\tau))^2 = \tau^{-2}(-m+n\tau)^2$. For (b) use $\eta(-1/\tau) = \sqrt{\tau/i}\,\eta(\tau)$ with $\eta(\tau) = q^{1/12}\prod(1-q^{2n})$, $q = e^{\pi i\tau}$. By Proposition 3.2 of Chapter 5, the logarithmic derivative in τ is

$$\frac{\eta'}{\eta}(\tau) = \frac{\pi i}{12} - 2\pi i\sum_{n=1}^{\infty}\frac{nq^{2n}}{1-q^{2n}}, \qquad \sum_{n=1}^{\infty}\frac{nq^{2n}}{1-q^{2n}} = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} nq^{2nm} = \sum_{k=1}^{\infty}\sigma_1(k)q^{2k}.$$

Since $F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum_k \sigma_1(k) q^{2k}$, $\frac{\eta'}{\eta}(\tau) = \frac{i}{4\pi} F(\tau)$. By the chain rule the logarithmic derivative of $\eta(-1/\tau)$ is $\tau^{-2}\frac{\eta'}{\eta}(-1/\tau)$, which by (a) equals $\frac{i}{4\pi}\tilde{F}(\tau)$. Taking logarithmic derivatives in the functional equation for η,

$$\frac{i}{4\pi}\tilde{F}(\tau) = \frac{1}{2\tau} + \frac{i}{4\pi}F(\tau),$$

so $\tilde{F}(\tau) = F(\tau) - 2\pi i/\tau$.

(c) follows from (a) and (b). ∎

Proof of Proposition 3.8. (ii) By (10) and Lemma 3.9(c),

$$\begin{aligned} E_2^*(-1/\tau) &= F(-1/(2\tau)) - 4F(-2/\tau) = [4\tau^2 F(2\tau) - 4\pi i\tau] - 4[(\tau/2)^2 F(\tau/2) - \pi i\tau] \\ &= 4\tau^2 F(2\tau) - \tau^2 F(\tau/2) = -\tau^2(F(\tau/2) - 4F(2\tau)) = -\tau^2 E_2^*(\tau). \end{aligned}$$

(iii) In $F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum \sigma_1(k) e^{2\pi i k\tau}$ the sum tends to 0 as $\operatorname{Im}(\tau) \to \infty$, so by (10) $E_2^*(\tau) \to \frac{\pi^2}{3} - 4\frac{\pi^2}{3} = -\pi^2$.

(iv) We first show

$$E_2^*(1 - 1/\tau) = \tau^2\Big[F\Big(\frac{\tau - 1}{2}\Big) - F(\tau/2)\Big]. \tag{11}$$

F has period 1, and Lemma 3.9(c) can be written $F(w) = w^{-2}[F(-1/w) + 2\pi i w]$. Let $u = (\tau - 1)/2$ and $w = u/\tau = \frac{1}{2} - \frac{1}{2\tau}$. Then $-1/w = -2 - \frac{2}{\tau - 1}$, so $F(-1/w) = F(-1/u) = u^2 F(u) - 2\pi i u$, and

$$F\Big(\frac{1}{2} - \frac{1}{2\tau}\Big) = \frac{\tau^2}{u^2}[u^2 F(u) - 2\pi i u + 2\pi i u/\tau] = \tau^2 F(u) - \frac{2\pi i \tau^2(1 - 1/\tau)}{u} = \tau^2 F\Big(\frac{\tau - 1}{2}\Big) - 4\pi i\tau.$$

Also $4F(2 - 2/\tau) = 4F(-1/(\tau/2)) = 4[(\tau/2)^2 F(\tau/2) - 2\pi i\tau/2] = \tau^2 F(\tau/2) - 4\pi i\tau$. Subtracting gives (11), since $E_2^*(1 - 1/\tau) = F(\frac{1}{2} - \frac{1}{2\tau}) - 4F(2 - \frac{2}{\tau})$. By the expansion of F, $F(\frac{\tau - 1}{2}) - F(\tau/2) = -8\pi^2 \sum_k \sigma_1(k) e^{\pi i k\tau}((-1)^k - 1) = 16\pi^2 e^{\pi i\tau} + O(|e^{3\pi i\tau}|)$, which with (11) gives (iv). For θ^4: (i)–(iii) follow from Section 1 ($\theta(-1/\tau)^4 = (\tau/i)^2\theta(\tau)^4 = -\tau^2\theta(\tau)^4$, $\theta \to 1$), and (iv) from Corollary 1.8: $\theta(1 - 1/\tau)^4 \sim 16(\tau/i)^2 e^{\pi i\tau}$. ∎

Proof of the four-squares theorem. Consider $f(\tau) = E_2^*(\tau)/\theta(\tau)^4$, holomorphic in $\mathbb{H}$, invariant under T_2 and S by Proposition 3.8. Since $\theta(\tau)^4 \to 1$ and $\theta(1 - 1/\tau)^4 \sim 16\tau^2 e^{\pi i\tau}(-1)$ as $\operatorname{Im}(\tau) \to \infty$, f is bounded in $\mathcal{F}$, hence in $\mathbb{H}$. By Theorem 3.4, f is constant, equal to $-\pi^2$ by (iii). So $\theta^4 = -\pi^{-2} E_2^*$, which proves the theorem; in particular $r_4(n) \geq 8 > 0$ for all $n \geq 1$. ∎